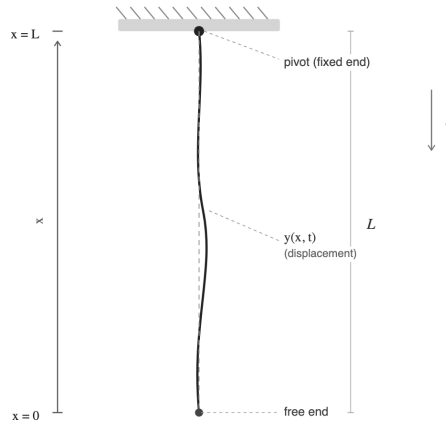


Becoming Best Friends with Bessel

1 *A Bit More on the Hanging String [Foundational]

In class last week, we discussed the normal modes of a string hanging under gravity. The tension is position dependent $T(x) = \mu gx$, and for a normal mode $y(x, t) = A(x)e^{i\omega t}$ of frequency ω , we found the following differential equation,

$$x \frac{d^2 A(x)}{dx^2} + \frac{dA(x)}{dx} + \frac{\omega^2}{g} A(x) = 0. \quad (1)$$



1. This is not a “standard” differential equation, and finding the allowed frequencies ω is not straightforward (unlike the simple string without gravity). In such cases, dimensional analysis comes in handy. The available physical parameters are mass density μ , length L and gravity g . Using dimensional analysis, what combination of these parameters can ω depend upon (up to some constant factor)? How does it compare with the simple pendulum?

Next, we will fill in the missing math from class to turn Eq. (1) into the standard Bessel equation form (of order α),

$$\frac{d^2 A}{du^2} + \frac{1}{u} \frac{dA}{du} + \left(1 - \frac{\alpha^2}{u^2}\right) A = 0. \quad (2)$$

2. Make the substitution $u = 2\omega\sqrt{x/g}$ and transform the all variables and derivatives into “ u -space”. For example,

$$\frac{d}{dx} = \frac{du}{dx} \frac{d}{du} = \frac{2\omega^2}{gu} \frac{d}{du}. \quad (3)$$

Do a similar transformation to get d^2/dx^2 in terms of u , d/du and d^2/du^2 .

3. Substitute everything back into Eq. (1) to show that our hanging string differential equation in u -space is nothing but the Bessel equation of order $\alpha = 0$,

$$\frac{d^2 A}{du^2} + \frac{1}{u} \frac{dA}{du} + A = 0. \quad (4)$$

2 *Why (Circular) Drums Sound “Thuddy” [Substantive]

Strike a guitar string and you hear a clear musical pitch. Its overtones are $2f_0, 3f_0, 4f_0, \dots$ — exact integer multiples of the fundamental — so the brain fuses them into a single, stable note. Strike a tabla and the pitch is elusive: a rich timbre, a characteristic thump, but no clean harmonic series. The reason is not craftsmanship or material. It is mathematics. The resonant frequencies of a circular membrane are not integer multiples of anything — they are irrational numbers set by the zeros of the Bessel functions $J_0, J_1, J_2, \dots$.

Consider a uniform circular membrane of radius R , surface mass density σ , held under tension $\mathcal{T}$ (force per unit length) and clamped at its rim. Let $g(r, \phi, t)$ be the small transverse displacement (Figure 1). The 2D wave equation in polar coordinates is

$$\frac{\partial^2 g}{\partial t^2} = v^2 \nabla^2 g = v^2 \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial g}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 g}{\partial \phi^2} \right], \quad v = \sqrt{\frac{\mathcal{T}}{\sigma}}. \quad (5)$$

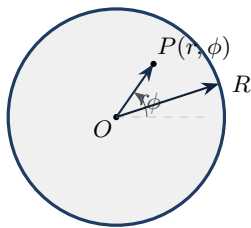


Figure 1: Top-view of the membrane in polar coordinates. The rim at $r = R$ is clamped: $g(R, \phi, t) = 0$.

- a. **Separation of variables.** Substitute the ansatz $g = R(r) \Phi(\phi) T(t)$ into (5) and divide through by $R\Phi T$.

- (i) Show that the time part separates to give $\ddot{T}/T = -\omega^2$, with solution $T(t) = A \cos \omega t + B \sin \omega t$.
- (ii) Show that the spatial remainder can be rearranged into the form

$$\frac{r^2}{R} \left(R'' + \frac{R'}{r} \right) + k^2 r^2 = -\frac{\Phi''}{\Phi} \equiv m^2, \quad k \equiv \frac{\omega}{v}.$$

Explain why both sides must equal the same constant m^2 , and write down the two resulting ODEs for R and Φ . (m here is the separation constant, not a mode index.)

- b. **Angular equation and quantisation of m .** The angular ODE $\Phi'' + m^2 \Phi = 0$ has general solution $\Phi = C \cos m\phi + D \sin m\phi$. Single-valuedness of g requires $\Phi(\phi + 2\pi) = \Phi(\phi)$.

- (i) Show that this forces m to be a non-negative integer: $m = 0, 1, 2, \dots$
- (ii) The integer m counts *nodal diameters* — straight lines through the centre on which $g = 0$. Sketch the nodal patterns for $m = 0, 1, 2$. (No calculation needed.)

- c. **Bessel’s equation.** With the substitution $u = kr$, show that the radial ODE becomes

$$\frac{d^2 R}{du^2} + \frac{1}{u} \frac{dR}{du} + \left(1 - \frac{m^2}{u^2} \right) R = 0. \quad (6)$$

This is **Bessel’s equation of order m** . Its two independent solutions are $J_m(u)$ and $Y_m(u)$; both are plotted in Figure 2.

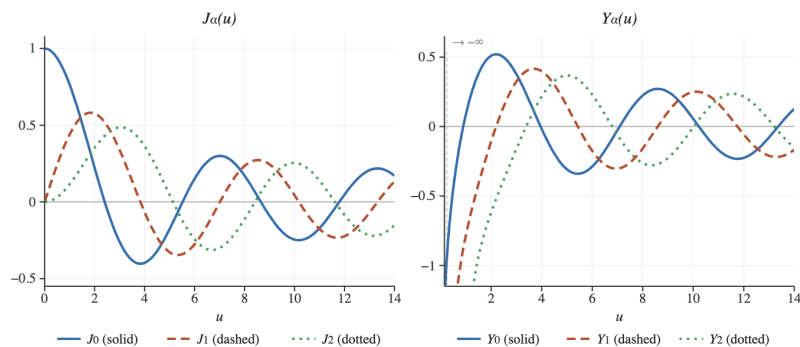


Figure 2: Bessel functions J_m (left) and Y_m (right) for $m = 0, 1, 2$. All J_m are finite at $u = 0$; all Y_m diverge there.

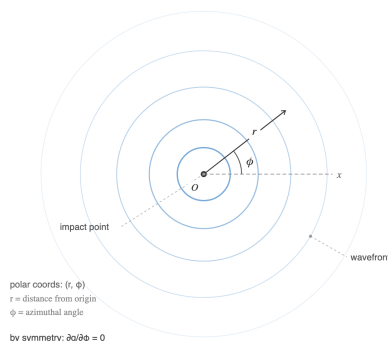
- (i) $Y_m(u) \rightarrow -\infty$ as $u \rightarrow 0$. Why must we discard Y_m for the drumhead? Write down the surviving radial solution $R(r)$.
- (ii) For $m = 0$, equation (6) reduces to $uR'' + R' + uR = 0$. Note that this is the same ODE encountered in the hanging-string problem (after a suitable substitution), confirming $R = J_0(kr)$ for $m = 0$ modes. Verify this by substituting $m = 0$ directly.
- d. **Eigenfrequencies.** The clamped boundary requires $g(R, \phi, t) = 0$, giving $J_m(kR) = 0$. Let α_{mn} denote the n -th positive zero of J_m :

	$n = 1$	$n = 2$	$n = 3$	$n = 4$
J_0 :	2.405	5.520	8.654	11.792
J_1 :	3.832	7.016	10.173	13.324
J_2 :	5.136	8.417	11.620	14.796

- (i) Show that $k_{mn} = \alpha_{mn}/R$ and hence derive the eigenfrequencies $\omega_{mn} = v\alpha_{mn}/R$. What does m count physically? What does n count?
- (ii) Compute the ratios $\omega_{01} : \omega_{11} : \omega_{21} : \omega_{02}$ (normalise to $\omega_{01} = 1$) and compare with the harmonic series $1 : 2 : 3 : 4$ of a stretched string. Are the overtones of a circular membrane harmonically related? Connect your answer to why a tabla sounds the way it does.

3 Ripples on a Pond: Amplitude Dependence in 2D [Substantive]

Drop a pebble into a still pond. A circular disturbance spreads outward from the point of impact — rings of crests and troughs expanding in all directions at the surface wave speed v . Far from the source, each wavefront is an ever-growing circle, and the amplitude of the ripple visibly diminishes as it spreads. But how exactly does it diminish? In this problem you will derive the answer from the wave equation itself, following precisely the same substitution strategy used in class for spherical waves in 3D — but now in two spatial dimensions.



The surface displacement $g(r, \phi, t)$ of a wave on a 2D surface satisfies the same wave equation (5) from Problem 2 (with v replaced by the appropriate surface wave speed).

(a) **Symmetry and simplification.** A pebble dropped at the origin creates a disturbance that, by symmetry, looks the same in every direction from the point of impact.

(i) What does this physical symmetry imply about the dependence of g on the angular coordinate ϕ ? Write down explicitly what $\partial g/\partial \phi$ and $\partial^2 g/\partial \phi^2$ must equal, and justify your answer in one sentence.

(ii) Using this, simplify the wave equation above to obtain the radially symmetric wave equation:

$$\frac{\partial^2 g}{\partial t^2} = \frac{v^2}{r} \frac{\partial}{\partial r} \left(r \frac{\partial g}{\partial r} \right).$$

(iii) We look for oscillatory solutions of the form $g(r, t) = A(r) e^{i\omega t}$. Substitute this ansatz into the radially symmetric equation from part (ii) and show that $A(r)$ satisfies the ODE for $k = \omega/v$.

$$\frac{1}{r} \frac{d}{dr} \left(r \frac{dA}{dr} \right) + k^2 A = 0,$$

(b) **The radial ODE is Bessel's equation.** Make the substitution $u = kr$ in the ODE for A from part (a)(iii). Show that it becomes:

$$\frac{d^2 A}{du^2} + \frac{1}{u} \frac{dA}{du} + A = 0,$$

or equivalently $\frac{d}{du} \left(u \frac{dA}{du} \right) + u A = 0$. This is *Bessel's equation of order zero*, whose two linearly independent solutions are $J_0(u)$ and $Y_0(u)$. Hence the exact solution is $A(r) = c_1 J_0(kr) + c_2 Y_0(kr)$. Why must we set $c_2 = 0$ for a physical solution? (Hint: what does $Y_0(u)$ do as $u \rightarrow 0$?)

(c) **The substitution.** Inspired by the 3D trick from class, try the substitution

$$A(r) = \frac{f(r)}{\sqrt{r}}$$

for some unknown function $f(r)$. Show that the ODE for $A(r)$ becomes, in terms of f :

$$f'' + \left(k^2 - \frac{1}{4r^2} \right) f = 0,$$

where primes denote derivatives with respect to r .

Careful: The $1/4r^2$ term does *not* vanish. This is the key difference from the 3D case, where the analogous substitution gave $f'' + k^2 f = 0$ exactly. In 2D, the geometry is less forgiving.

(d) **Far-field approximation.** For large r (far from the source, i.e. $kr \gg 1$), the term $1/4r^2$ is negligible compared to k^2 . Drop the $1/4r^2$ term and solve the resulting ODE for $f(r)$. Write down the two linearly independent solutions. Hence write down the approximate far-field form of $A(r)$, and thereby the full solution $g(r, t) = A(r) e^{i\omega t}$, keeping the outgoing wave (expanding away from the source).

(e) **Amplitude decay.** From your result in part (d), read off how the amplitude of the ripple depends on r . Compare with the 3D result from class. Fill in the table:

Dimension	Wavefront geometry	Amplitude $\propto$
1D (string)	flat (plane)	constant
2D (pond surface)	circle	?
3D (sound, light)	?	$1/r$

(f) **Energy argument.** Confirm your answer using a simple energy conservation argument: the total energy carried by the wave is spread over the wavefront. How does the size of the wavefront (its length in 2D, its surface area in 3D) grow with r ? Since intensity $I \propto g^2$ and the total power crossing any wavefront is constant, how must g scale with r in each case? Does this agree with your calculation?